

LDV_GSL

Gasoline Light Duty Vehicle



Gasoline Car

Inputs / Outputs

dir	comm	unit	group
Input	GSL	PJ	i
Output	PLDVHWY	MPKm	o
Output	PLDVCTY	MPKm	o

Efficiency (ceff)

	comm	use2cact	cact2cout	share.up
2	PLDVHWY	679.995	2.03	0.4
3	PLDVCTY	502.401	2.03	0.8

Key Parameters

cap2act	12.04
olife	16 years
stock	22948 1000 Vehicles
cap unit	1000 Vehicles
act unit	million km, city
cost unit	USD/(Vh*km)

Fuel Efficiency

Mode	Fuel	L/100 km	km/kWh	MJ/km	mpg (US)
Highway	GSL	4.30	2.45	1.47	54.70
City	GSL	5.82	1.81	1.99	40.41

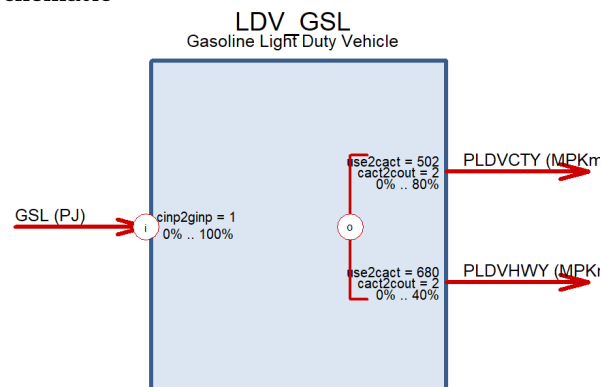
Vehicle Parameters

Fuels	GSL	Primary fuel	GSL
Efficiency (highway)	4.3 L/100km	Efficiency (city)	5.82 L/100km
Annual distance	12,043 km/yr	Capacity unit	1000 vehicles
Load/vehicle (hwy)	2.03 passengers	Load/vehicle (city)	2.03 passengers
Max share (highway)	40%	Max share (city)	80%
Vehicle cost	8,414 USD	Annual fixed cost	25 USD/yr
Lifetime	16 yr		

Capacity bounds

region	year	stock	cap.lo	cap.up	cap.fx	ncap.lo	ncap.up	ncap.fx	ret.lo	ret.up	ret.fx
IND	2024	22948.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2025	21513.8	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2030	14342.5	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2035	7171.2	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2040	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2045	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
IND	2050	0.0	NA	NA	NA	NA	NA	NA	NA	NA	NA

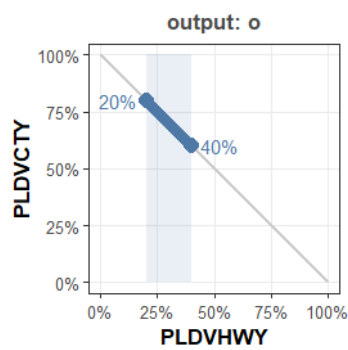
Schematic



Costs [USD/(Vh*km)]

Fixed O&M	0.02
Investment cost	8.41

Share Mix Constraints



Levelized Cost 0.17 [USD/(Vh*km)], 0.08 [USD/(Passenger*km)]

